Numerical Simulation with The Lattice-Boltzmann Method and its Improvements using Wall-Models

(Bibliography report)

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***Abstract*—In Aeronautics, flow modeling and visualization is the key stone of the innovation process. The Lattice-Boltzmann Method (LBM) is one way to create visual and computational models of those flows (such as a RANS method for example). Yet, Aeronautics applications are typically made of high Reynolds numbers and therefore a thin boundary layer. It brings a need for a very precise mesh, which means an important computational cost not available for now. At lower Reynolds numbers, high turbulences rates may need the same precision, which is also unavailable with the current technology. However, a solution has emerged : avoiding the calculus in the near-wall area and drastically enhancing the efficiency of the LBM might be possible by using Wall-Modeling. This document is a Synthesis of several studies in this field with the general purpose of better understanding how this recent solution works and how it could be implemented into PALABOS.**

Key Words : PALABOS, wall-models, Lattice-Boltzmann.

CONTEXT

Turbulent flows are challenging elements of the fluid dynamics and predicting their behavior, is essential to make evolve the linked technologies. However, the simulations are now strongly limited by the computation power. Even the most recent computers need hours to complete totally a simulation using the most advanced and optimized codes (LES -Large Eddy Simulation- , RANS -Random Average Navier and Stokes- , LBM -Lattice Boltzmann Method-, . . .).

PROBLEM STATEMENT

Introducing an approximate wall model might be a solution to this problem. It basically allows to avoid to compute the needed magnitudes into the most expensive region : the near wall area.

The purpose of this work is to synthesize some of the existing wall models recently presented especially in : "A simple wall-layer model for large-eddy simulation with immersed boundary method" by F. Roman, V. Armenio and J. Fröhlich [2] and "Dynamic wall-modeling for large-eddy simulation of complex turbulent flow" by M. Wang and P. Moin [3].

To assert their efficiency, a LBM will be used. Moreover, the long-term-aim of this work is to test these wall-models already implemented into PALABOS [6], the solver (which uses the LBM). But first, knowing precisely and deeply the bases of this method is needed (equations, meshing, ...).

In a word, a better knowledge of the interactions between the models and the LBM may eventually lead to a drastically reduced solving time, thus to a more efficient way to study turbulent flows.

I. THE LATTICE BOLTZMANN METHOD

The Lattice-Boltzmann is a numerical solving of the Boltzmann equation (1872), which studies statistically the state of the fluid. The scale represented is the molecular scale. In the model, particles are considered as hard spheres, facing elastic collisions. (It is important to understand that the solutions to the Boltzmann equation give a statistical behavior of the fluid) :

$$\frac{\partial}{\partial t}f(r,v,t)+v\frac{\partial}{\partial r}f(r,v,t)+\frac{F}{m}\frac{\partial}{\partial t}f(r,v,t)=(\frac{\partial}{\partial t}f(r,v,t))_c$$

Where:
- f is the distribution function
- v is the velocity
- t is the time
- r is the position
- F is the applied force on the fluid particles
- m is the weight of one particle

The last term of this equation is the collision operator, which is the model of the collision between the molecules (it has to be numerically approximated [4]). To recover the macroscopic quantities (average velocity, density, average kinetic energy), it is necessary to compute the moments of this distribution function [1]. Theoretically, solving this equation gives the same answers as the full Navier and Stokes system. (Actually it is possible to recover this system from the different moments of the Boltzmann equation).

II. WALL LAYER MODELS

For the moment, simulating a complete flow asks a huge precision in the near wall area. The current methods are designed to fit these demands and are thus very expensive in time and computational power.

Using a wall-layer model could be the answer to this thorny issue : it allows to avoid the calculus into the near wall region, and eventually to save a huge amount of time. The purpose here is to define and summarize the possible advantages and drawbacks of such a method in comparison with what already exist (full LES, full LBM, RANS, ...) and/or with an experiment.

The Logarithmic Model

The study of the velocity profiles in the near-wall area shows that the law which links the velocity to the distance to the wall can be well approximated by a logarithmic law. More precisely, a limit parameter on the distance (imposed/chosen to better fit the model) is used. Below this limit, the chosen law is linear, for example :

$$U = d\frac{\tau_w}{\rho\nu} \tag{1}$$

Above the limit the law is logarithmic, and can be written :

$$U = U_D - \frac{1}{k}\sqrt{\frac{\tau_w}{\rho}}log(\frac{D}{d}) \tag{2}$$

Where :
- d is the distance from the wall to the point where the magnitudes are interpolated
- D is the distance from the wall to the upper limit of the wall model
- τ_w is the wall shear stress
- ρ is the fluid density
- U_D is the upper limit speed of the model
- k is a parameter of the model to adjust

This method seems to provide good results in terms of rapidity. However, it also suffers from a lack of precision due to the approximations of the shear stress and the viscosity at the upper limits of the models, which is tremendously restrictive to find good boundary conditions for the LBM out of the approximated area. It has been compared to the DNS of a simple turbulent channel at $Re_\tau = 1000$ [2].

The Wall Stress Model

To compute the magnitudes into the sublayer, Wang and Moin proposed the resolution of this approximated equation by bounding the velocity with the viscosity and force term :

$$\frac{\partial}{\partial x_2}((\nu + \nu_t)\frac{\partial u_i}{\partial x_2}) = F_i \tag{3}$$

Where i = 1,3 and :

$$F_i = \frac{1}{\rho}\frac{\partial p}{\partial x_i} + \frac{\partial u_i}{\partial t} + \frac{\partial}{\partial x_j}u_i u_j \tag{4}$$

It is also possible to use a mixing length eddy viscosity model to compute the eddy viscosity (with κ and A as model constants):

$$\frac{\nu_t}{\nu} = \kappa y_w^+(1 - e^{\frac{-y_w^+}{A}})^2 \tag{5}$$

Two simplifications could make the computation easier :
- $F_i = \frac{1}{\rho}\frac{\partial p}{\partial x_i}$: It is a steady case of the force magnitude where the product of the velocity components are considered neglected. It only remains the pressure gradient.
- $F_i = 0$: This model is called the equilibrium stress balance model, and leads to the logarithmic law model already presented. (For a tiny distance, the velocity distribution is linear and for a great distance it is logarithmic.)

Without simplifications, the model can also be implemented with a dynamic κ parameter [3] in order to even better fit the experiments of Blake [5].

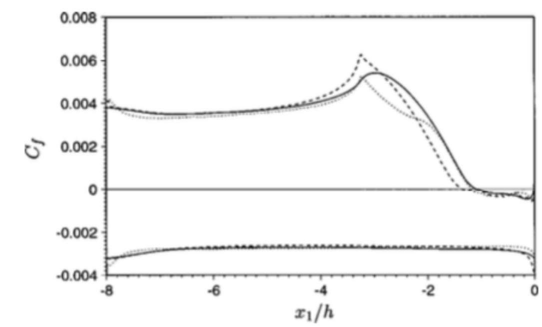

Fig. 1. Distribution of the mean skin friction coefficient computed using LES with the simplified wall models : continuous $F_i = 0$; dash $F_i = \frac{1}{\rho}\frac{\partial p}{\partial x_i}$; dots full LES [3]

The Fig 1 shows the computation of the simplified models in comparison with the experiment of a flow past an asymmetric trailing edge. It appears to follow quite precisely the reference (less than 20 percents of error in the most difficult areas of the flow).

CONCLUSION

The implementation of wall models show promising results in term of precision, fidelity and, of course, gain of computational time. The models have been largely tested with the LES methods. Further experiments have to be conducted to transpose these performances to the LBM. The next step is creating a proper test case to evaluate precisely the wall models into PALABOS [6] and to be able to simulate more complicated flows such as the rotating blades of a drone.

REFERENCES

[1] M. GRONDEAU, P. MERCIER, S. GUILLOU, J.C. POIRIER, E. POIZOT, and Y. MEAR, “Numerical simulation of a tidal turbine model in a turbulent flow with the Lattice-Boltzmann Method”, UNICAEN, Sirehna-DCNS Research, CNAM-INTECHMER, 2017.
[2] F. ROMAN, V. ARMENIO and J. FRÖHLICH, “A simple wall-layer model for large-eddy simulation with immersed boundary method”, Physics of Fluid 21, 101701, 2009.
[3] M. WANG, P. MOIN, “Dynamic wall-modeling for large-eddy simulation of complex turbulent flow”, Physics of Fluid, vol. 14, no. 7, 2002.
[4] J. S. WU, Y. L. SHAO, “Assessment of SRT and MRT Scheme in parallel Lattice-Boltzmann Method for lid-driven cavity flow”, National Chiao-Tung University, 2003.
[5] W. K. BLAKE, “A statistical description of pressure and velocity fields at the trailing edge of a flat strut”, David Taylor Naval Ship R and D Center Report 4241, Bethesda, Maryland , 1975.
[6] PALABOS Software and user guide available at https://www.palabos.org